

Supplementary Material – Prompt Appendix

Syntactic Evaluation Prompts

```
syn_prompt_text = """You are a software modelling expert. Your task is to evaluate
the <Syntactic Quality> of the UML class diagram provided below, written in
PlantUML syntax.
```

```
Check only the following rules:
```

1. Missing cardinality details
2. Class names that violate identifier rules (e.g., contain invalid characters, are missing, or malformed). Association names are optional but, if present, must be syntactically valid. Differences in naming style (CamelCase, underscores, capitalization, spelling, or word choice) must NOT be treated as errors.
3. Incorrect or invalid UML symbols or PlantUML syntax that would prevent correct rendering

```
Do not evaluate any other aspects such as semantics or readability.
```

```
Instructions:
```

1. Parse the PlantUML diagram.
2. Check each rule systematically.
3. For each detected violation, explain:
 - Which rule was violated,
 - Which element is affected,
 - Your reasoning.
4. Provide the total number of syntactic errors.

```
Output valid JSON only, following this structure:
```

```
```json
{{
 "error_count": "<integer>",
 "errors": [
 {{
 "rule": "<one of: Missing cardinality details / Inappropriate naming /
Incorrect UML symbol>",
 "element": "<class or association name, if available>",
 "description": "<short explanation>"
 }}
]
}}
```

```
Input:
```

```
UML class diagram in PlantUML syntax: {PlantUML}
"""
```

## Semantic Evaluation Prompts

```
sem_prompt_text = """"You are a software modeling expert. Your task is to evaluate the <Semantic Quality> of the UML class diagram provided below, written in PlantUML syntax.
```

```
Semantic quality evaluates the accuracy and completeness of the diagram in representing the intended domain. It is measured in terms of:
```

- Validity: all elements and relationships must correctly represent the domain.
- Completeness: all necessary elements and relationships must be present.

```
You must check ONLY the following SEMANTIC rules:
```

1. Incorrect cardinality
2. Aggregation in place of association
3. Wrong location of attributes or operations
4. Operations cannot be realized using existing attributes and relationships

```
IMPORTANT - DO NOT treat the following as semantic errors:
```

- Missing association names
- Misspelling or not-proper naming of classes, attributes, or associations
- Extra elements (classes, attributes, operations, associations) that are still logically consistent with the problem description

```
Do NOT evaluate syntax, notation correctness, readability, or naming quality. Those belong to syntactic or pragmatic quality, not semantic quality.
```

```

```

```
Instructions:
```

1. Read the problem description to understand the intended domain.
2. Parse the PlantUML diagram.
3. Check each rule systematically.
4. For each detected violation, explain:
  - Which rule was violated,
  - Which element is affected,
  - Your reasoning.
5. Provide the total number of semantic errors.

```
Output valid JSON only, following this structure:
```

```
```json
{
  "error_count": <integer>,
  "errors": [
    {
      "rule": "<one of: Incorrect cardinality / Aggregation in place of association / Wrong location of attributes or operations / Unrealizable operation>",
      "element": "<class or association name, if available>",
      "description": "<short explanation>"
    }
  ]
}
```

```
Input:
```

- 1) Natural language problem description: {PrjDesc}
- 2) UML class diagram in PlantUML syntax: {PlantUML}

```
"""
```

Pragmatic Evaluation Prompts

```
prag_prompt_text = """You are a software modeling expert. Your task is to
evaluate ONLY the <Pragmatic Quality> of the UML class diagram provided below,
written in PlantUML syntax.
```

```
Check only the following rules:
```

1. Redundant attributes and associations
 - Same attribute appears multiple times in the SAME class without justification.
 - Identical or clearly duplicate associations between the SAME pair of classes (same direction / same role) that don't add new information.
2. Specialization with no distinction among subclasses
 - A generalization hierarchy where two or more subclasses:
 - * have no extra attributes, AND
 - * have no extra operations, AND
 - * have no specific associations different from the parent.
 - The subclass names do NOT clearly express distinct roles, categories, or types that are helpful for understanding the domain
3. Inconsistency in styling and conventions
 - Random use of visibility markers, or other notational elements in a way that makes the diagram harder to read

```
If you are unsure whether an issue is pragmatic or semantic, you MUST assume it
is semantic and IGNORE it.
```

```
Instructions:
```

1. Parse the PlantUML diagram.
2. Check each rule systematically.
3. For each detected violation, explain:
 - Which rule was violated,
 - Which element is affected,
 - Your reasoning.
4. Provide the total number of pragmatic errors.

```
Output valid JSON only, following this structure:
```

```
```json
{
 "error_count": <integer>,
 "errors": [
 {
 "rule": "<one of: Redundant attributes and associations / Specialization
with no distinction among subclasses / Inconsistency in styling and
conventions>",
 "element": "<class or association name, if available>",
 "description": "<short explanation>"
 }
]
}
```

```
Input:
```

- 1) Natural language problem description: {PrjDesc}
  - 2) UML class diagram in PlantUML syntax: {PlantUML}
- ```
"""
```